Smart Public Transport Analytics
Recommendations

Teyzix Core - Data Analytics Internship - Task DA-3

Overview

This document presents actionable recommendations for the transport authority, derived from the analysis of 1,300 trips across eight routes. Each recommendation follows directly from a specific insight identified in the analysis, and is intended to improve reliability, reduce cost, and enhance the passenger experience.

1. Route Optimization

1. **Review low-revenue routes (R06, R02).** Assess whether frequency can be reduced or smaller buses assigned, reallocating vehicles to higher-demand corridors without cutting service where it is genuinely needed.

2. Schedule Improvements

2. **Add weather-contingent buffer times.** Since delays rise roughly 75% in fog, rain, and storms, published schedules should include automatic buffer minutes during poor weather to keep timetables realistic and improve reliability.

3. Peak Hour Fleet Planning

3. **Concentrate capacity at 7–8 AM and 5–6 PM.** Deploy the largest, highest-capacity buses during these windows and consider reducing mid-day and weekend frequency, where occupancy is lower.

4. Revenue Optimization

4. **Protect and expand high-fare routes.** Routes like R01 (airport) drive disproportionate revenue; ensuring reliable, frequent service on premium corridors protects the strongest income stream. Explore modest fare or service upgrades on high-demand, low-fare routes.

5. Delay Reduction Strategies

5. **Target weather and congestion, not routes.** Because delays are uniform across routes but spike with weather and at rush hour, invest in real-time traffic routing and peak-hour priority measures rather than route-specific fixes.

6. Fuel Efficiency Improvements

6. **Expand the Electric fleet.** Electric buses use roughly one-tenth the fuel per passenger of other types. Gradually replacing Mini and older diesel buses with Electric vehicles would cut operating costs and emissions substantially.

7. Passenger Experience Enhancements

7. **Reduce cancellations and crowding.** Address the 8.3% cancellation rate (especially on R05 and R08) through better vehicle and driver scheduling, and relieve peak-hour crowding with the added capacity from recommendation 3 to improve rider satisfaction.

Conclusion

The main opportunities for this transport operation are operational: manage delays through weather-aware scheduling, concentrate fleet capacity at predictable commuter peaks, expand the highly efficient Electric fleet, and reduce cancellations. Acting on these data-driven recommendations would improve reliability, lower costs, and enhance the passenger experience without major structural change.